V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning

META

July 8, 2026

Seowon Lee

Introduction

Previous SSL for Video Representation

- Goal:
  - Learn high-level representations from raw image/videos
  - Capture both global scene understanding and local spatio-temporal information

Introduction

Previous SSL for Video Representation

- Limitations:
  - JEPA / V-JEPA : strong global representation but weak dense/local features
  - Dense SSL (e.g., DINO) : good spatial correspondence but limited temporal modeling

Challenge: Learn representation that preserve

both dense spatio-temporal structure and **global semantics**

Introduction

V-JEPA 2.1 propose:

1) Dense Predictive Loss
2) Deep Self-Supervision
3) Multi-modal Tokenizer
4) Scaling

- Goal: Improve **dense video representation** while maintaining strong global understanding

Introduction

- Goal: Improve dense video representation while maintaining strong global understanding

Preliminaries: JEPA

JEPA (Joint-Embedding Predictive Architecture)

- Key Idea:
  - SSL framework
  - Predict latent representations instead of pixels
  - Learn by predicting masked tokens

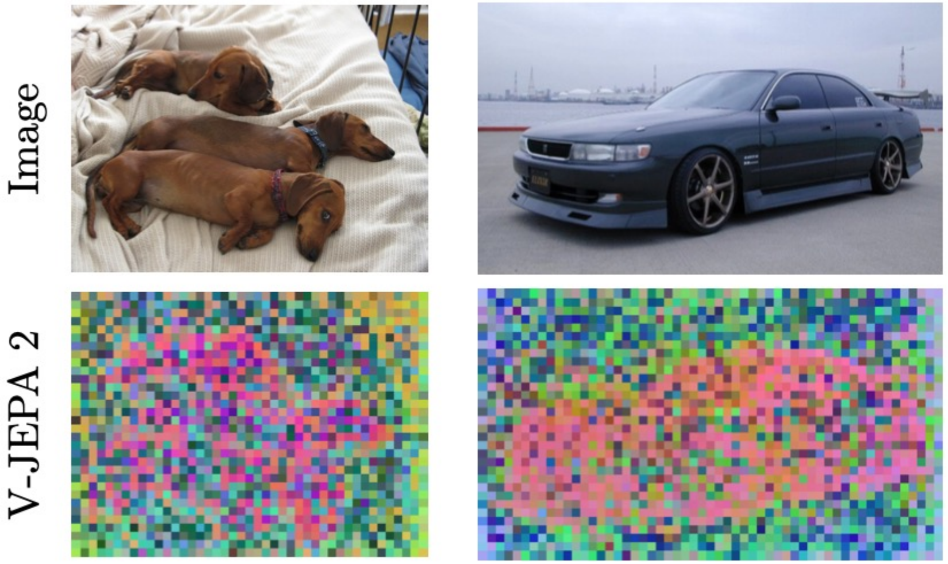
- Prediction loss on **masked tokens only**

$$\underset{\theta,\phi,\Delta_y}{\text{minimize}} \left\| P_\phi\left(\Delta_y, E_\theta(x)\right) - \text{sg}\left(E_{\overline{\theta}}(y)\right) \right\|_1$$

Preliminaries: JEPA

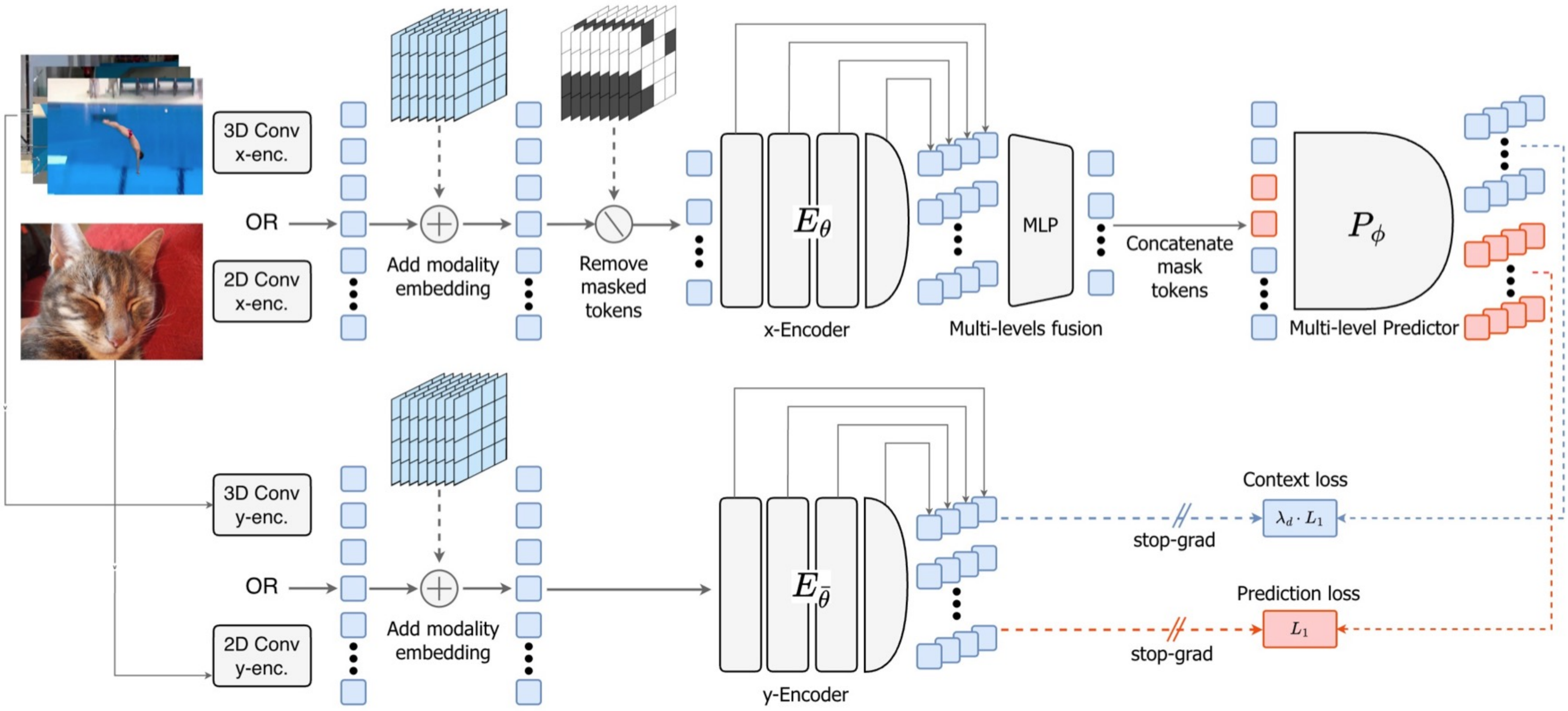
Limitations of JEPA

- Observation
  - Good global representation
  - Weak local spatial structure
  - Poor dense tasks performance
    - Semantic segmentation / depth estimation



- Hypothesis: Lack of self-supervision on patches that are **NOT masked**
  - context tokens become global aggregators
  - Leads to weak dense features
  - **V-JEPA 2.1: Context Self-Supervision**

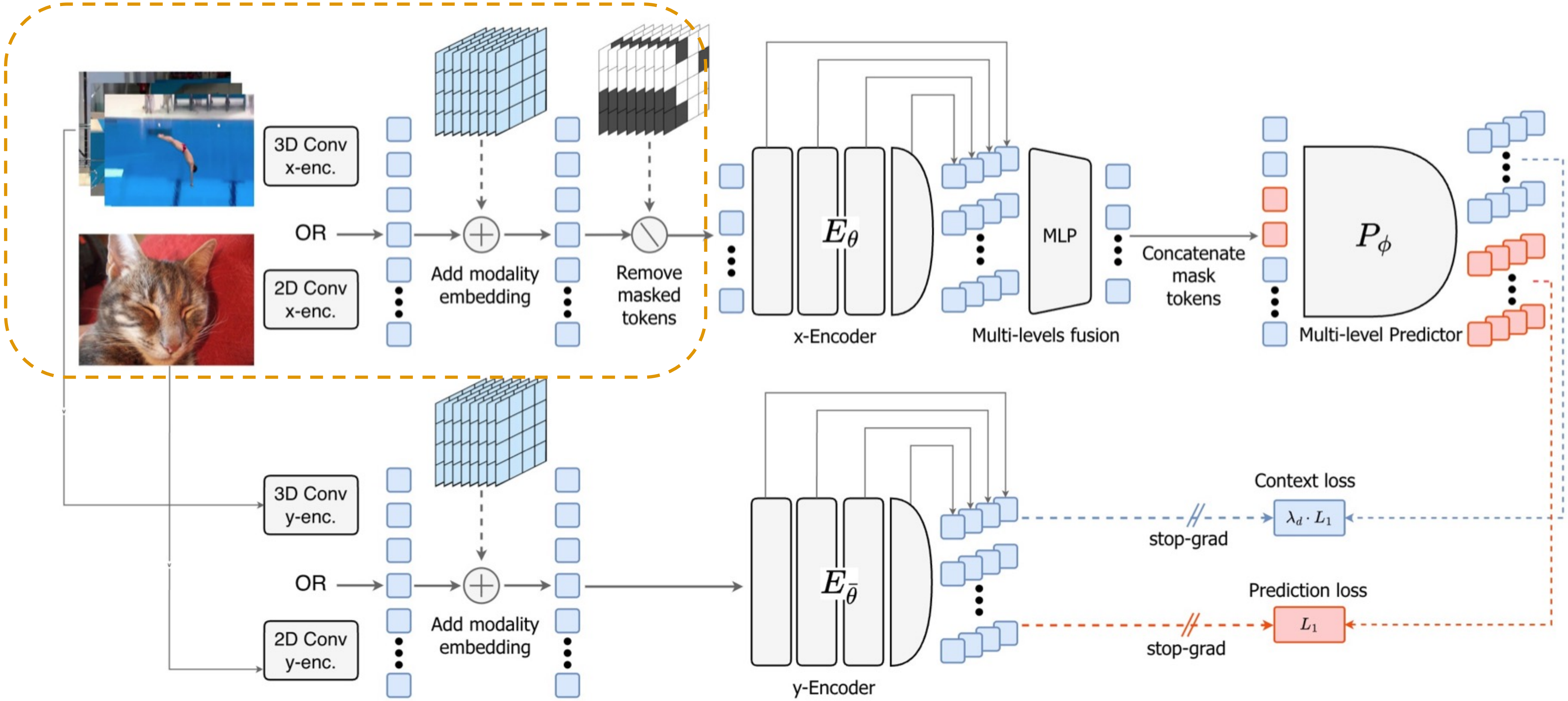
Architecture of V-JEPA 2.1



Architecture of V-JEPA 2.1

(1) Input Processing

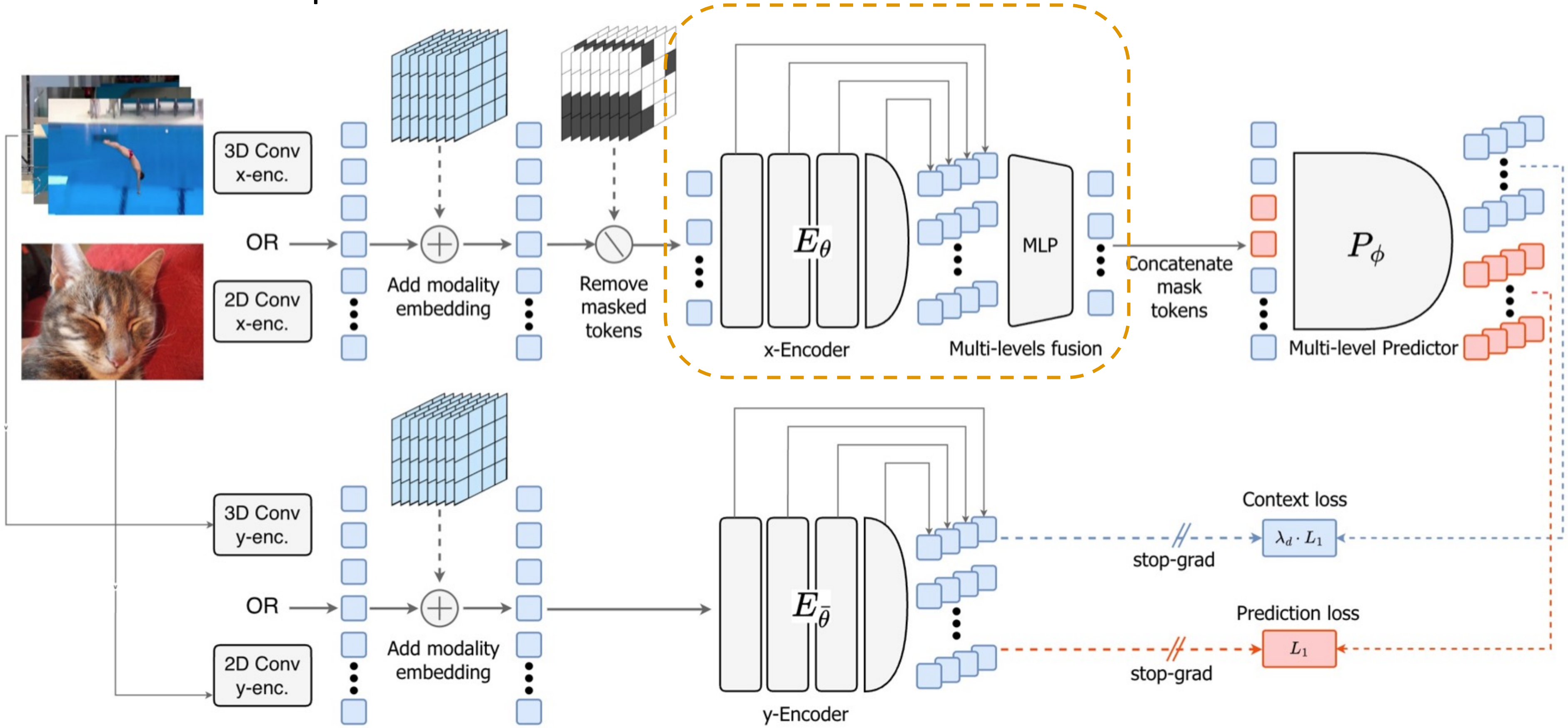
- Modality-specific patch embedding
- Randomly mask patch tokens



Architecture of V-JEPA 2.1

(2) X-Encoder & Multi-level Fusion

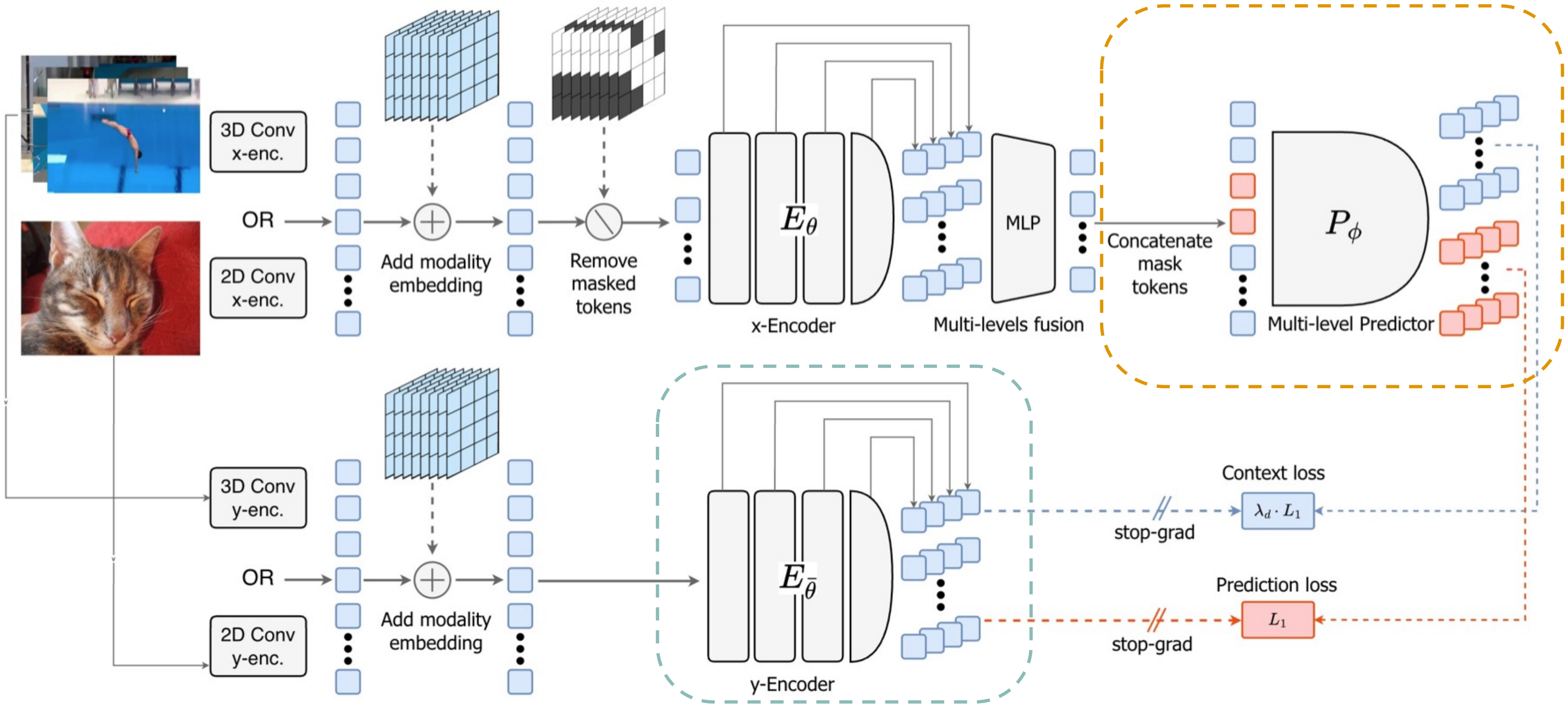
- X-Encoder extracts **multi-level** representations
- Concatenate (**Intermediate features** + Final output) along channel axis
- MLP → Fused context representation



Architecture of V-JEPA 2.1

(3) Predictor

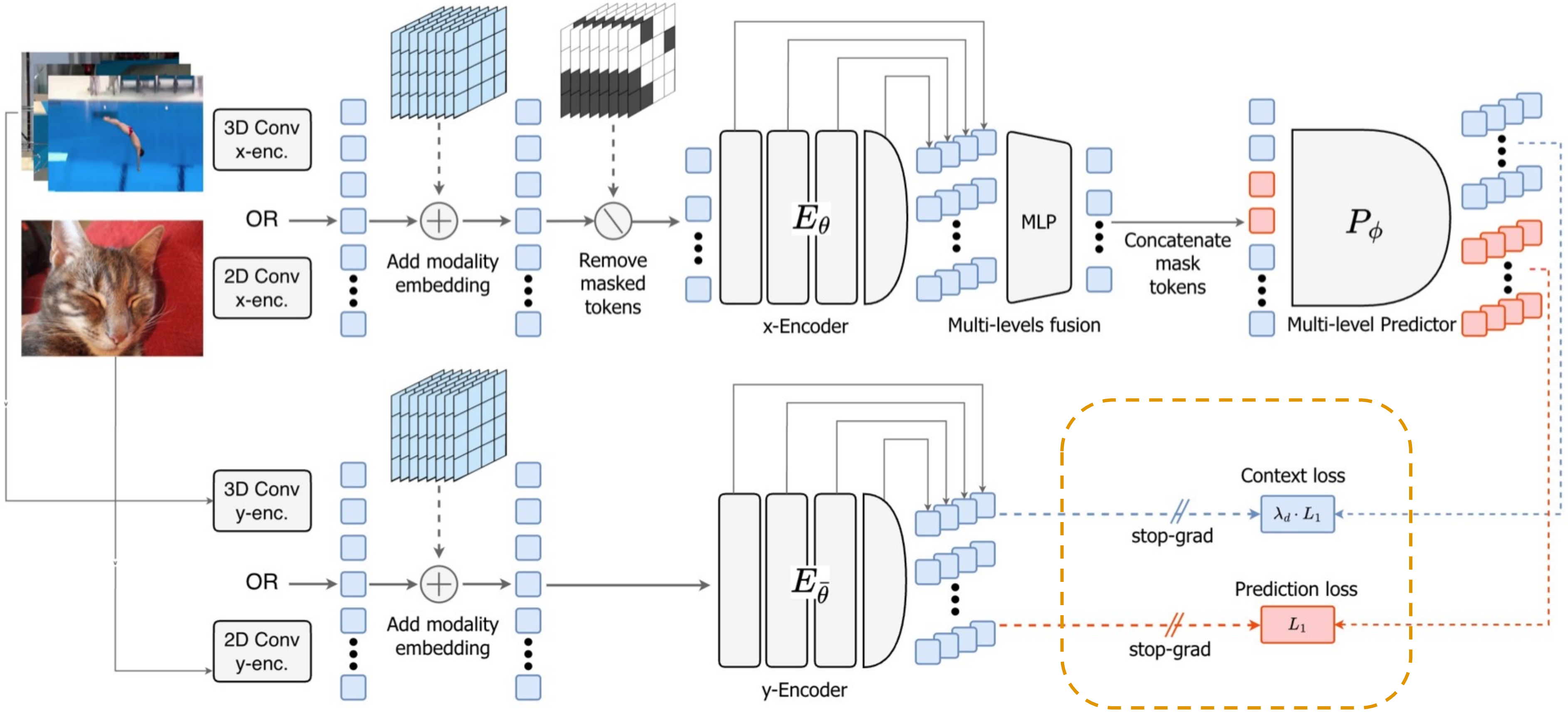
- Concatenate context tokens with mask tokens
- Predict representations for **all tokens**



Architecture of V-JEPA 2.1

(4) Training Objective

- Prediction Loss (masked tokens) + **Context Loss (visible tokens)**
- Minimize $L_{predict} + L_{ctx}$



Dense Prediction Loss

Context Self-Supervision: minimize $L_{predict} + L_{ctx}$

Prediction Loss (V-JEPA 2) :

$$\mathcal{L}_{predict} = \frac{1}{|M|}\sum_{i \in M} \left\| P_\phi\left(E_\theta(x), \Delta_y\right)_i - \mathrm{sg}(E_{\overline{\theta}}(y)_i) \right\|_1$$

M : set of masked patch

Context Loss (V-JEPA 2.1) :

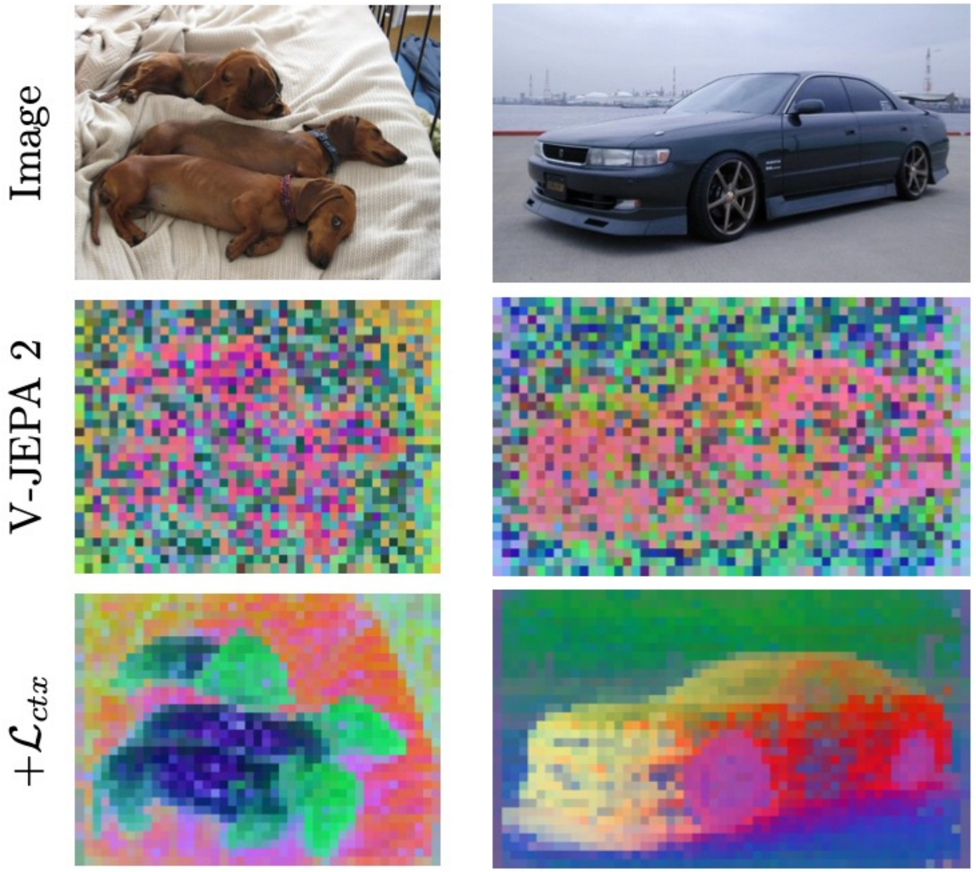
$$\mathcal{L}_{context} = \frac{1}{|C|}\sum_{i \in C} \lambda_i \left\| P_\phi\left(E_\theta(x), \Delta_y\right)_i - \mathrm{sg}(E_{\overline{\theta}}(y)_i) \right\|_1$$

C : set of context token

λ : patch-specific weight

Dense Prediction Loss

Context Self-Supervision



IN1K Acc.	SSv2 Acc.	NYU RMSE ↓	ADE20K mIoU	
82.2	72.8	0.682	22.2	V-JEPA 2
72.6	62.5	0.474	33.8	+ Context Loss

Global Task :

- Image Classification (IN1K)
- Video Classification (SSv2)

Dense Task:

- Depth Estimation (NYUv2)
- Semantic Segmentation (ADE20K)

- Poor performance on global semantic tasks

Dense Prediction Loss

Weighting Coefficients

(1) Fixed Weight

- Same λ
- Dense ↑ / Global ↓

(2) Warm-up

- Gradually increase λ
- Stable training

(3) Dynamic Weighting

ADE20K mIoU	SSv2 Acc.	λ	Scheme	Warmup
22.2	72.8	V-JEPA 2, $\lambda = 0$	cte	
26.4	71.0	0.05	cte	
29.6	62.5	0.2	cte	
27.5	53.8	0.5	cte	
24.6	51.1	1.0	cte	
30.5	60.5	0.2	cte	✓
32.2	61.5	0.5	cte	✓
33.8	62.5	0.5	weighted	✓

Dense Prediction Loss

Weighting Coefficients

(1) Fixed Weight

- Same λ
- Dense ↑ / Global ↓

(2) Warm-up

- Gradually increase λ
- Stable training

(3) Dynamic Weighting

ADE20K mIoU	SSv2 Acc.	λ	Scheme	Warmup
22.2	72.8	V-JEPA 2, $\lambda = 0$	cte	
26.4	71.0	0.05	cte	
29.6	62.5	0.2	cte	
27.5	53.8	0.5	cte	
24.6	51.1	1.0	cte	
30.5	60.5	0.2	cte	✓
32.2	61.5	0.5	cte	✓
33.8	62.5	0.5	weighted	✓

Dense Prediction Loss

Weighting Coefficients

(3) Dynamic Weighting

- Patch-specific λ (distance aware)
- Best dense/global trade-off

$$\lambda_i = \frac{\lambda}{\sqrt{d_{\min}(i, M)}}$$

d_{min} : distance to the nearest masked token

- **Near** masked patch $\longrightarrow$ Larger weight
- **Far** from mask $\longrightarrow$ Smaller weight

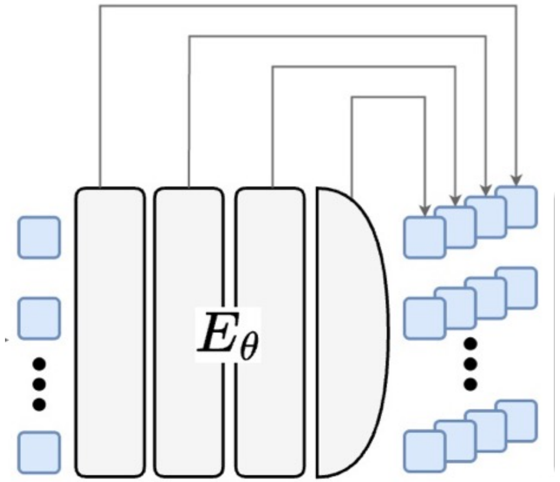
ADE20K mIoU	SSv2 Acc.	λ	Scheme	Warmup
22.2	72.8	V-JEPA 2, $\lambda = 0$	cte	
26.4	71.0	0.05	cte	
29.6	62.5	0.2	cte	
27.5	53.8	0.5	cte	
24.6	51.1	1.0	cte	
30.5	60.5	0.2	cte	✓
32.2	61.5	0.5	cte	✓
33.8	62.5	0.5	weighted	✓

- Produces smoother, less noisy feature maps

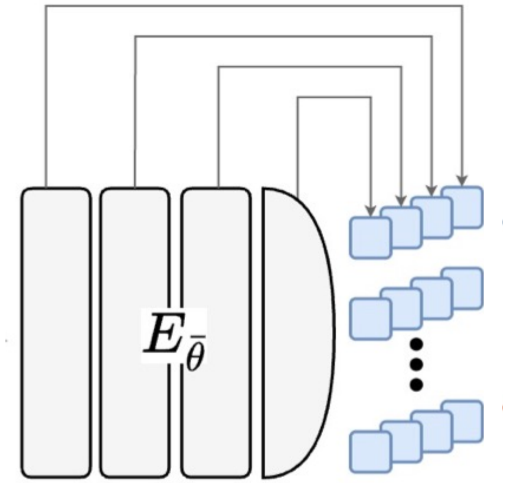
Deep Self-Supervision

Multi-level Representation Supervision

- Limitation of V-JEPA 2
  - Supervision is applied only to the final encoder output
  - Intermediate layers are not explicitly optimized
- Proposed:
  - Loss is applied to all representation levels
- Effect:
  - Allow local information to flow toward final layer
  - Improve dense + global downstream tasks



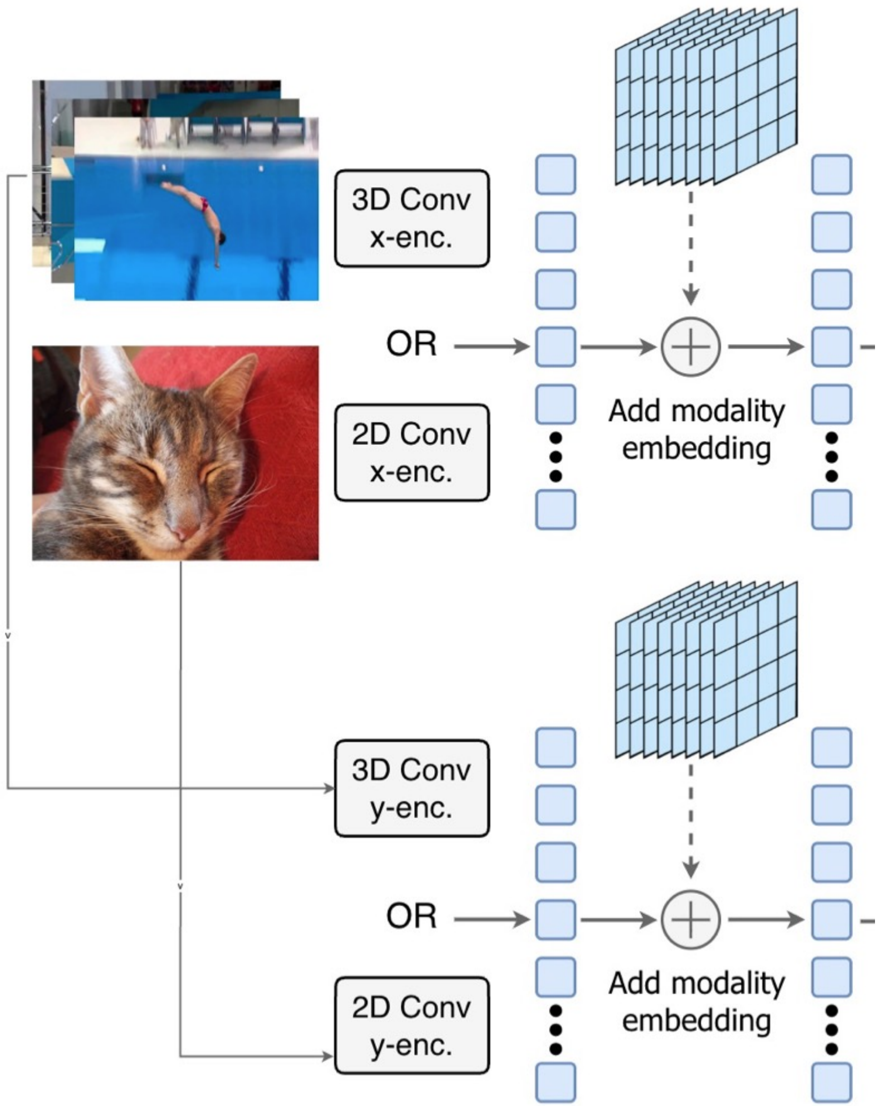
x-Encoder



y-Encoder

Multimodal Tokenizer

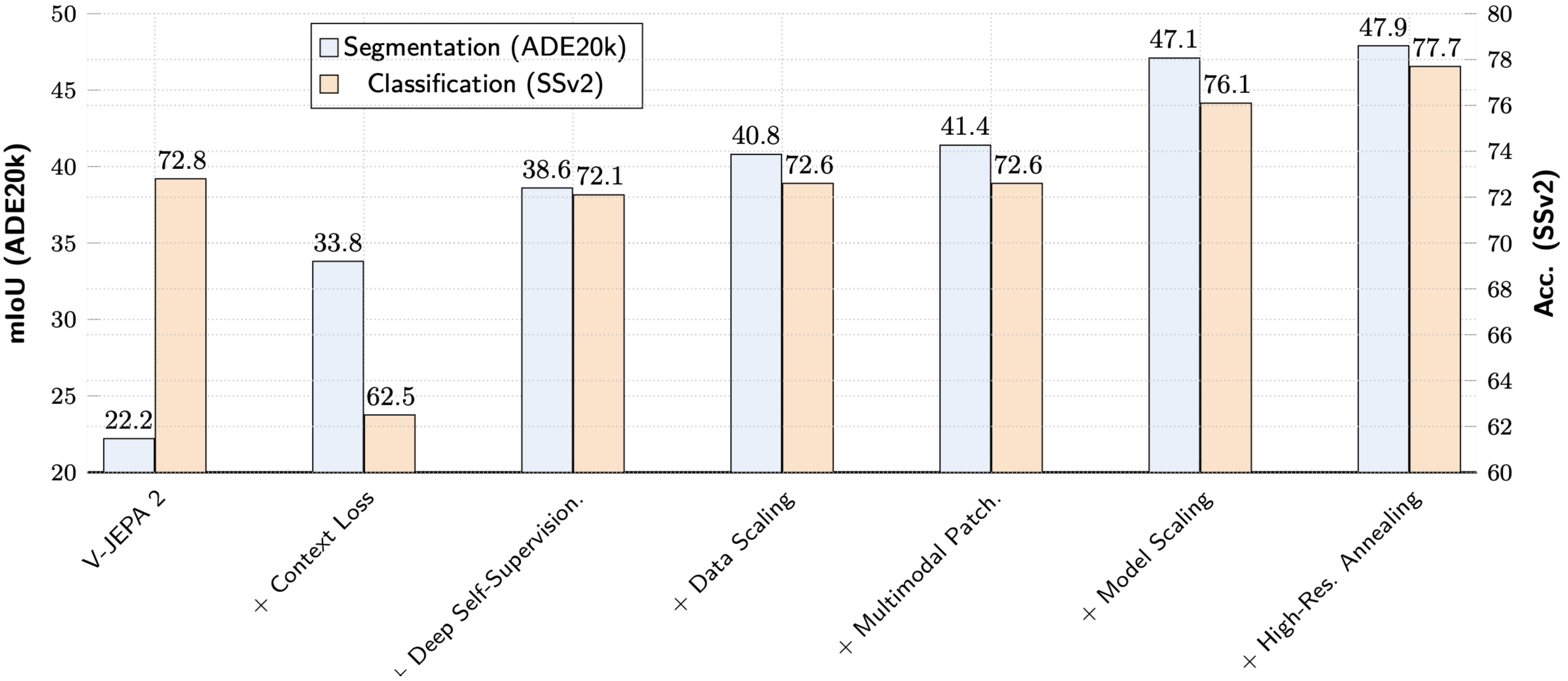
- Previous V-JEPA 2:
  - Single 3D convolution tokenizer
  - Images treated as 16-frame static videos
- Problem:
  - Unnecessary computation
  - Temporal bias for images
- V-JEPA 2.1:
  - Image: 2D Conv
  - Video: 3D Conv
  - Add modality learnable Token



Scaling

- Data Scaling
  - Construct VisionMix163M
  - Large-scale curated image + video sources
  - Increase dynamic/motion-rich video samples
- Model Scaling
  - Encoder Scaling: ViT-L (300M) $\longrightarrow$ ViT-G (2B)
  - High-resolution Cool-down

Impact of V-JEPA 2.1 Design Components



Conclusion

Key Contributions

- Deep spatio-temporal self-supervision
  - Weighted context loss + multi-level supervision
  - Improves dense local features
- Scaling training recipe
  - Modality-specific tokenizer
  - Large-scale data + model scaling

Key Result:

- Achieves strong dense + global representations